NGUYEN NGOC THANH

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EDUCATION

Lawrence S. Ting School (Dinh Thien Ly Secondary and High School) | Ho Chi Minh City **2023 - 2027**

GPA: Grade 9 - **9.6/10** Grade 10 - **9.5/10** Grade 11 - **9.6/10**

Standardized Testing: SAT: **1490 (Math: 800)** IELTS **7.5** AP Biology: **5** AP Calculus BC: **5** AP Statistics: **4**

AWARD & HONORS

- **Top 5% Academic Standing** | Lawrence S. Ting School | Four Consecutive Years | 2023–2026
- **Lawrence S. Ting School Academic Scholarship** | Four Consecutive Years | 2023 - 2026
- **Gold Award** | Philippine International Math and Science Olympics (PIMSO) | Dec 2025
- **Gold Award** | IMOCSEA–ISOCSEA National Round, Science (Grade 12) | 2026
- **Most Exemplary Student of House Fantasy** | 1 of 4 school-wide recipients across 1,000+ students | 2025
- **Gold** | StemCo 2025 Summer Chemistry Competition, Grades 9-10 | 2025
- **Silver** | StemCo 2025 Summer Biology Competition, Grades 9-10 | 2025
- **Exemplary Young Leader** | School recognition through application and faculty review | 2026
- **Honorable Mention** | DONMUN, Special Political & Decolonization Committee | 2026
- **Third Prize** | Mathematics, 30/4 Olympic Competition | 2025
- **Student with Excellent Demonstration of Six Core Values** | 2023, 2026

RESEARCH

Computer Vision & Medical AI Research

First Author & Student Researcher **Nov. 2023 - Present**

Advisor: *Bach-Tung Pham, Ph.D. | Department of Computer Science and Information Engineering, National Central University, Taiwan*

Current manuscript: *Photometric Augmentation and Appearance Regularization for Self-Supervised Skin-Lesion Classification* – submitted to Computer Methods and Programs in Biomedicine, Oct. 2026

- Progressed from emotion recognition to medical AI: built FER-2013 models and a real-time inference prototype (2023–24), conducted a 36-run image-augmentation study (2024), then transitioned to ISIC melanoma classification and self-supervised learning (2025–present).
- Adapted Barlow Twins into an end-to-end dermoscopic-image pipeline and designed six conditions testing Planckian Jitter and appearance regularization across 401,059 unlabeled and 33,569 labeled downstream images.
- Conducted 200+ full training experiments (~600 GPU-hours); the best current single-seed model increased independent-test F1 from 0.5522 to 0.5795 (+4.9% relative) while showing an AUROC trade-off; further validation is ongoing.

Biological Control of Phytophthora palmivora in Durian Using Bacillus subtilis - Lam Ha, Lam Dong

Team Lead & Primary Researcher, 2-student team **May 2024 - Present**

Advisor: *Lê Phước Thạnh, Ph.D., Lecturer & Researcher, Department of Plant Protection, College of Agriculture, Can Tho University*

Research Mentor: *Trương Phúc Thịnh, B.Sc. in Plant Protection, Can Tho University*

- Initiated and led a field and laboratory study investigating whether *Bacillus subtilis* could inhibit *Phytophthora palmivora* in durian.
- Conducted field sampling and repeated pathogen-isolation attempts using symptomatic durian plant material collected in Lam Ha.
- Revised the research workflow in response to inconsistent isolation results and seasonal constraints, prioritizing reproducibility before further testing.
- Selected as 1 of ~15 students for the 2025 International Science Exchange Program; delivered a 6-minute English presentation on the preliminary study to visiting partner schools from Taiwan.

LEADERSHIP EXPERIENCE

Hoa Yen Charity Fund

Founder & President **Apr. 2024 - Present**

- Created a student-led community initiative from ~15 to 25 core members, coordinating 20–25 volunteers per event across youth volunteers, university students, community partners, and local businesses.
- Led fundraising, program planning, partnerships, and on-site execution, with approximately VND 284 million in documented program expenditures/support and 330+ documented beneficiary participations.
- Expanded Hoa Yen from a single water-safety campaign into recurring programs supporting children, elderly residents, and rural communities, with documented service across Can Tho and Lam Ha, Lam Dong.
  - ***Cai Rang Floating Market*:** Distributed 100 life jackets while promoting water-safety awareness in a river-dependent community.
  - ***Children & Youth:*** Organized a waste-for-gifts initiative and Mid-Autumn programs serving 100+ children in 2024 and approximately 150 children in 2026; supported *Chạm đến ước mơ* and Mai Tam Shelter for children living with HIV.
  - ***Rural Community Support:*** Coordinated rice distribution, rural-lighting support for 50+ households, and gifts for elderly residents as Hoa Yen expanded its work beyond Can Tho in 2025.
  - ***Elderly Care:*** Built a $1,200 care fund for elderly people living alone in Can Tho, combining fundraising with direct community visits and support in 2024 and 2025.

MoleBio - Molecular Biology Club, Lawrence S. Ting School

Founder & President, MoleBio **2025 - 2026**

- Relaunched a 16-member, lecture-heavy club as a 20-member organization centered on hands-on science.
- Designed 36 weekly sessions totaling 48 hours of programming, including laboratory safety, pipetting, centrifugation, DNA extraction/purification, PCR, and gel electrophoresis.
- Organized an osmosis and microscopy workshop that received 60+ registrations for 25 lab spots.
- Led a 7-member Science Fair team in a terrarium-building activity for 30 participants using recycled plastic bottles and locally collected materials.
- Initiated student-led projects, including a club website prototype and a physical 3D DNA model.

Member, Predecessor Molecular Biology Club **2024 - 2025**

- Identified limitations in the club's lecture-heavy format and lack of hands-on learning opportunities.
- Drafted a redesign plan incorporating laboratory activities, community outreach, and student-led projects.

Family Representative Council, Lawrence S. Ting School

F08 Representative; Documentation Committee **May 2024 - Present**

- Elected annually as 1 of 32 representatives serving a high-school community of 1,000+ students across four Houses.
- Collected and synthesized student feedback on facilities, well-being, sustainability, and school life for discussions.
- Proposed online feedback forms to broaden student input beyond representative-only reporting.
- Contributed to a student well-being/onboarding framework; Council feedback also helped prompt facility improvements, including swimming-pool cleaning and shared-seating maintenance.
- Helped draft FRC's vision and mission, defining priorities around student representation, well-being, and engagement.

INTERNSHIP

Stem Cell Summer School 2024 - Stem Cell Institute, University of Science, Vietnam National University Ho Chi Minh City

Research Trainee **Jul. 8 - Aug. 12, 2024**

- Analyzed research topics in stem cell biology through research-group seminars and scientific discussions.
- Contributed to research planning and data analysis under researcher mentorship.
- Conducted laboratory experiments to investigate biological questions and apply concepts in stem cell research.

Molecular Biology Course - Lawrence S. Ting School

Research Trainee **Jun. 16 - 20, 2025**

- Applied molecular biology and gene technology concepts through an intensive five-day wet-lab course.
- Performed DNA extraction and PCR, followed by gel electrophoresis to examine molecular samples.
- Analyzed molecular biology data and explored phylogenetic relationships through tree-based analysis.

Binh Phu Pharmaceutical Trading & Services Co., Ltd.

Laboratory Assistant Intern **Sep 2025**

- Assisted with pharmaceutical formulation under laboratory supervision and gained hands-on exposure to drug preparation techniques.
- Conducted product quality tests, including tablet friability, hardness, and moisture content.
- Developed foundational knowledge of pharmaceutical manufacturing processes, drug identification, and quality control procedures.

MUSIC, ARTS & PERFORMANCE

Co-writer & Performer **- Fantasy House Family Performing Arts - Tao Quan** **2025**

- Co-wrote and performed in a Family production before ~280 students and teachers; placed 3rd of 8 Families.

Team Lead, Writer, Director & Performer **| "The Beauty of Vietnamese Culture" Project - Cao Van Lau Musical** **Jan. 2025**

- Led a 6-student team to First Prize in a grade-wide project involving ~300 students; wrote and directed a 15-minute musical dramatizing composer Cao Van Lau's life and performed an excerpt from *Da co hoai lang*, including a cai luong duet.

Main Antagonist **- Fantasy House Performing Arts - "Hoa Chi"** **Nov. 2025**

- Selected through audition for the principal antagonist role in a 50–60-person House production after 1+ month of rehearsal; performed before 1,000+ students, teachers, and staff as Fantasy House placed 1st of 4 Houses.

Main Antagonist (Cast) **- Khoi Sac 27 - Annual Lawrence S. Ting School Performing Arts Fundraiser | Sep. 2026 - May 2027**

- Selected through two audition rounds for the main antagonist role in the school's annual student-led performing-arts fundraiser; currently rehearsing for the May 2027 production.

OTHER ACTIVITIES

Leadership Program **- The Academy at Harvard Square & Harvard Student Agencies** **Jul. 6 - 25, 2025**

- Completed a three-week experiential program combining academic visits and a business project at Harvard and MIT, cultural immersion and outdoor activities in New Hampshire, and educational travel through New York City and Washington, D.C

SKILLS & INTERESTS

- **Programming & AI:** Python; TensorFlow/Keras; computer vision; CNNs; self-supervised learning; model training/evaluation
- **Molecular Biology & Lab:** DNA extraction/purification; PCR; gel electrophoresis; microscopy; pipetting; centrifugation
- **Research:** Literature review; experimental design; data analysis; scientific presentation; field sampling; troubleshooting; reproducibility-focused research